



Chapter 3: The Phoenix Lands

Python

Students begin to see positive impacts of the Darkness being defeated, but must go to the last star well. Students use data types, loops, boolean methods, & nested conditionals.

Table of Contents

- 3. Pinefield
- 4. Forest Pace
- 5. Watch Out Tengshe
- 6. Ready or Not
- 7. A New Companion
- 12. A Dog Named Mouse
- 14. Mouse's First Trick
- 15. Doggie Diversion
- 19. Six Minus
- 20. Distancing Yourself
- 21. Broken Line
- 22. Far Away Bridge
- 25. Irrigation
- 29. Race for Water
- 30. Water Stairs
- 33. Lighting the Way
- 34. Incrementing Paths
- 35. Spiral Streets
- 39. Through Together
- 40. Dig, Jump, Move, Repeat
- 41. Loopstraction
- 42. Long Step Stair

43. Plus Minus Bridges
44. Eye of the Storm
48. For Mouse
54. For Inside
55. Triple Slalom
56. Serpentine
57. Haven Stairs
58. Dark Downstairs
60. Blind Steps
61. A Loop in the Fog
64. Double Choice
65. Into the Unknown
69. Storm Rising
70. Creeping Through the Tall Grass
71. Out of the Loop
74. Right Door
82. Knock Knock
83. Patience
84. While Twins
85. Shadow Behind
90. Be Slow, Be Quick
91. One Step Up
92. Charge the Time
95. Lost in Time
99. Time Call
100. Time Swap
101. Tempus Fugit
102. Tempus Impetus
110. Dangerous Skies

#3. Pinefield

Level Overview and Solutions

Pinefield

Intro

We may have trapped the Darkness for now, but these lands are still in turmoil.

Default Code

```
1 hero.moveRight()
```

Pinefield Solution

```
1 hero.moveRight()  
2 hero.moveUp(2)  
3 hero.sneakUp(2)  
4 hero.moveUp()  
5 hero.jumpRight()  
6 hero.moveRight()  
7 hero.moveDown()  
8 hero.jumpRight()  
9 hero.moveUp()
```

#4. Forest Pace

Level Overview and Solutions

Forest Pace

Intro

Oh no, there are more of them! Get past the Tengshe so we can reach the Sun Palace.

Default Code

```
1 hero.sneakUp()
```

Forest Pace Solution

```
1 hero.sneakUp()  
2 hero.sneakRight(2)  
3 hero.moveRight(1)  
4 hero.sneakRight(3)  
5 hero.jumpUp()  
6 hero.moveUp(1)  
7 hero.moveLeft(3)  
8 hero.moveUp(1)
```

#5. Watch Out Tengshe

Level Overview and Solutions

Watch Out Tengshe

Intro

You found some totems! Now you can use some spells to get past these enemies.

Default Code

```
1 enemy = hero.findNearestEnemy()
2 distance = hero.getDistanceTo(enemy)
3 if distance < 3:
4     hero.castIllusionMist(enemy)
5     hero.sneakUp(2)
6 else:
7     hero.moveUp(2)
```

Watch Out Tengshe Solution

```
1 enemy = hero.findNearestEnemy()
2 distance = hero.getDistanceTo(enemy)
3 if distance < 3:
4     hero.castIllusionMist(enemy)
5     hero.sneakUp(2)
6 else:
7     hero.moveUp(2)
8
9 hero.moveUp(3)
10 hero.moveRight(1)
11
12 enemy = hero.findNearestEnemy()
13 distance = hero.getDistanceTo(enemy)
14 if distance < 3:
15     hero.castIllusionMist(enemy)
16     hero.sneakRight(2)
17 else:
18     hero.moveRight(2)
19
20 hero.moveRight(2)
21 hero.moveDown(2)
22
23 enemy = hero.findNearestEnemy()
24 distance = hero.getDistanceTo(enemy)
25 if distance < 3:
26     hero.castIllusionMist(enemy)
27     hero.sneakDown(2)
28 else:
29     hero.moveDown(2)
30
31 hero.moveDown(1)
```

#6. Ready or Not

Level Overview and Solutions

Ready or Not

Intro

You can use your totems to sneak past the enemy. But don't use too many!

Default Code

```
1 enemy = hero.findNearestEnemy()
2 distance = hero.getDistanceTo(enemy)
```

Ready or Not Solution

```
1 enemy = hero.findNearestEnemy()
2 distance = hero.getDistanceTo(enemy)
3
4 if distance < 3:
5     hero.castEarthWave()
6
7 hero.moveUp(2)
8
9 hero.jumpUp()
10 hero.moveUp()
11 hero.jumpRight()
12 hero.moveRight(3)
13 hero.moveDown()
14
15 enemy = hero.findNearestEnemy()
16 distance = hero.getDistanceTo(enemy)
17
18 if distance < 3:
19     hero.castIllusionMist(enemy)
20     hero.sneakDown(2)
21 else:
22     hero.moveDown(2)
23
24 hero.jumpDown()
```


#7. A New Companion

Level Overview and Solutions

A New Companion

Intro

Isn't that Octans from the Moon Dancers? Looks like he's trapped!

Default Code

```
1 hero.moveRight(1)
```

A New Companion Solution

```
1 hero.moveRight(1)
2 hero.castEarthWave()
3 hero.moveRight(2)
4 hero.jumpRight()
5 enemy = hero.findNearestEnemy()
6 hero.castIllusionMist(enemy)
7 hero.sneakUp(3)
8 hero.use("bell")
9 hero.moveRight(2)
```

#12. A Dog Named Mouse

Level Overview and Solutions

A Dog Named Mouse

Intro

You and Mouse got separated. Time for some teamwork!

Default Code

```
1 hero.moveRight()  
2 mouse.moveRight()  
3 mouse.moveUp(2)  
4 mouse.moveRight(3)
```

A Dog Named Mouse Solution

```
1 hero.moveRight()  
2 mouse.moveRight()  
3 mouse.moveUp(2)  
4 mouse.moveRight(3)  
5  
6 # end of sample code  
7  
8 hero.moveRight(2)  
9 mouse.moveDown(2)  
10 mouse.moveRight(2)  
11 hero.moveRight(2)  
12 mouse.moveDown()
```

#14. Mouse's First Trick

Level Overview and Solutions

Mouse's First Trick

Intro

We need to cross the river. The buttons trigger the drawbridges, but you'll need Mouse's help.

Default Code

```
1 mouse.digLeft()  
2 mouse.digUp()
```

Mouse's First Trick Solution

```
1 mouse.digLeft()  
2 mouse.digUp()  
3 mouse.moveLeft()  
4  
5 hero.moveUp(5)  
6 hero.moveRight(1)  
7  
8 mouse.moveRight()  
9 mouse.digDown()  
10 mouse.digRight()  
11 mouse.moveRight(3)  
12 mouse.moveUp(5)
```

#15. Doggie Diversion

Level Overview and Solutions

Doggie Diversion

Intro

Bzzzz... Sorry, I seem to be glitching more. There are - BZZT - creatures ahead. You can use Mouse to distract them so you can sneak past. Bzzt...

Default Code

```
1 mouse.moveRight(2)
2 mouse.bark()
```

Doggie Diversion Solution

```
1 mouse.moveRight(2)
2 mouse.bark()
3
4 hero.sneakRight(4)
5 hero.jumpUp()
6 hero.jumpRight()
7
8 mouse.moveRight(2)
9 mouse.moveDown()
10 mouse.moveRight(3)
11
12 hero.moveRight(1)
```

#19. Six Minus

Level Overview and Solutions

Six Minus

Intro

Vega is malfunctioning, but I, Octans, can help you! Just let me check my notes...

Default Code

```
1 birdPerson = hero.findNearestEnemy()
2 stepsToBird = hero.getDistanceTo(birdPerson)
3
4 mouse.moveRight(stepsToBird)
5 mouse.bark()
6
7 remainingSteps = 6 - stepsToBird
```

Six Minus Solution

```
1 birdPerson = hero.findNearestEnemy()
2 stepsToBird = hero.getDistanceTo(birdPerson)
3 remainingSteps = 6 - stepsToBird
4 mouse.moveRight(stepsToBird)
5 mouse.bark()
6
7 hero.sneakRight(5)
8 hero.moveUp(1)
9 hero.jumpUp()
10
11 mouse.moveRight(remainingSteps)
12 mouse.moveUp(5)
13 hero.moveUp(1)
```

#20. Distancing Yourself

Level Overview and Solutions

Distancing Yourself

Intro

I can almost see Haven on the horizon!

Default Code

```
1 enemy = hero.findNearestEnemy()  
2 steps = hero.getDistanceTo(enemy)
```

Distancing Yourself Solution

```
1 enemy = hero.findNearestEnemy()  
2 steps = hero.getDistanceTo(enemy)  
3 mouse.moveRight(steps)  
4 mouse.bark()  
5  
6 hero.sneakRight(5)  
7 hero.sneakUp(1)  
8 hero.jumpUp()  
9 hero.jumpRight()  
10 mouse.moveRight(5 - steps)  
11 mouse.moveDown(1)  
12 mouse.moveRight(2)  
13 mouse.moveUp()  
14 mouse.digUp()
```

#21. Broken Line

Level Overview and Solutions

Broken Line

Intro

I can almost see Haven on the horizon! We just need to get past one more Bird Person.

Default Code

```
1 enemy = hero.findNearestEnemy()  
2 steps = hero.getDistanceTo(enemy)
```

Broken Line Solution

```
1 enemy = hero.findNearestEnemy()  
2 steps = hero.getDistanceTo(enemy)  
3  
4 mouse.moveUp(steps)  
5 mouse.bark()  
6  
7 mouse.moveRight(2)  
8 mouse.digRight()  
9 remainingSteps = 5 - steps  
10 mouse.moveUp(remainingSteps)  
11  
12 hero.moveRight(1)  
13 hero.jumpRight()  
14 hero.jumpRight()
```

#22. Far Away Bridge

Level Overview and Solutions

Far Away Bridge

Intro

There's the gate to Haven! But I didn't see this coming... there are **more** Bird People in the way!

Default Code

```
1 maxSteps = 7
```

Far Away Bridge Solution

```
1 maxSteps = 7
2
3 nearestBird = hero.findNearestEnemy()
4 mouseSteps = hero.getDistanceTo(nearestBird)
5
6 mouse.moveRight(mouseSteps)
7 mouse.bark()
8 remainingSteps = maxSteps - mouseSteps
9 mouse.moveUp(2)
10 mouse.moveRight(remainingSteps)
11 mouse.digUp()
12
13 hero.moveUp()
14 hero.jumpUp()
15 hero.jumpRight()
16 hero.jumpUp()
17 hero.sneakRight(4)
```

#25. Irrigation

Level Overview and Solutions

Irrigation

Intro

Haven was known as a green city, full of lovely gardens! But the time storm caused debris to block the water, and now everyone is so sad!

Default Code

```
1 hero.moveUp()  
2 hero.castEarthRoll("right")
```

Irrigation Solution

```
1 hero.moveUp()  
2 hero.castEarthRoll("right")  
3  
4 hero.moveUp()  
5 hero.castEarthRoll("right")  
6  
7 hero.moveUp()  
8 hero.castEarthRoll("right")  
9  
10 hero.moveUp()  
11 hero.castEarthRoll("right")  
12  
13 hero.moveUp()  
14 hero.castEarthRoll("right")
```

#29. Race for Water

Level Overview and Solutions

Race for Water

Intro

You're doing great! And with **for loops**, you should be able to do even better!

Default Code

```
1 for i in range(5):  
2     hero.moveDown()
```

Race for Water Solution

```
1 for i in range(5):  
2     hero.moveDown()  
3     hero.castEarthRoll("right")
```

#30. Water Stairs

Level Overview and Solutions

Water Stairs

Intro

We're almost done! And with those **for loops**, we should be able to fix all the waterways in no time!

Default Code

```
1 for i in range(4):  
2     hero.moveRight()
```

Water Stairs Solution

```
1 for i in range(4):  
2     hero.moveRight()  
3     hero.moveUp()  
4     hero.castEarthRoll("right")
```

#33. Lighting the Way

Level Overview and Solutions

Lighting the Way

Intro

All the **torches** in the town were put out by the winds of the storm. Give us light again!

Default Code

```
1 torchName = "torch"
2 walkDistance = 1
3
4 for i in range(3):
5     hero.moveRight()
6     hero.moveUp(walkDistance)
7     # what do I do next?
```

Lighting the Way Solution

```
1 torchName = "torch"
2 walkDistance = 1
3
4 for i in range(3):
5     hero.moveRight()
6     hero.moveUp(walkDistance)
7     hero.use(torchName)
8     hero.moveRight()
9     hero.moveDown(walkDistance)
10    walkDistance = walkDistance + 1
```

#34. Incrementing Paths

Level Overview and Solutions

Incrementing Paths

Intro

Okay, we got some more torches to light. You can use your for loop and those variable things to get it done in a snap!

Default Code

```
1 torchName = "torch"
2 walkDistance = 1
3
4 for i in range(3):
5     hero.use(torchName)
6     hero.moveRight()
7     # use walkDistance then increase it
```

Incrementing Paths Solution

```
1 torchName = "torch"
2 walkDistance = 1
3
4 for i in range(3):
5     hero.use(torchName)
6     hero.moveRight()
7     hero.moveUp(walkDistance)
8     hero.use(torchName)
9     hero.moveRight()
10    walkDistance = walkDistance + 1
11    hero.moveDown(walkDistance)
12    walkDistance = walkDistance + 1
```

#35. Spiral Streets

Level Overview and Solutions

Spiral Streets

Intro

This place was so beautiful before the time storm hit. And with your help, it can be beautiful again.

Default Code

```
1 walkDistance = 6
```

Spiral Streets Solution

```
1 walkDistance = 6
2
3 for i in range(3):
4     hero.moveRight(walkDistance)
5     hero.moveUp(walkDistance)
6     hero.use("torch")
7     walkDistance = walkDistance - 1
8     hero.moveLeft(walkDistance)
9     hero.moveDown(walkDistance)
10    walkDistance = walkDistance - 1
11    hero.use("torch")
```

#39. Through Together

Level Overview and Solutions

Through Together

Intro

Okay, time to go into that... scary... Time Rift thing. I'll stay back while you and Mouse go first...

Default Code

```
1 # Start with a loop for the hero actions
2
3
4 mouse.moveRight(5)|
```

Through Together Solution

```
1 for i in range(3):
2     hero.moveUp(2)
3     hero.moveRight()
4     hero.moveDown(2)
5     hero.moveRight()
6
7 mouse.moveRight(5)
8 mouse.moveDown(3)
9 mouse.moveRight()
10 hero.moveRight()|
```

#40. Dig, Jump, Move, Repeat

Level Overview and Solutions

Dig, Jump, Move, Repeat

Intro

Okay, so it looks like the Rift changes everything around us, but the path stays the same. You can still use your loops.

Default Code

```
1 for i in range(3):  
2     mouse.moveUp()  
3     mouse.digRight()
```

Dig, Jump, Move, Repeat Solution

```
1 for i in range(3):  
2     mouse.moveUp()  
3     mouse.digRight()  
4  
5  
6 for j in range(3):  
7     hero.moveDown(1)  
8     hero.jumpRight()  
9  
10  
11 mouse.moveDown(3)  
12 hero.moveRight()  
13 hero.moveUp(2)
```

#41. Loopstraction

Level Overview and Solutions

Loopstraction

Intro

So this Rift isn't too bad. You just need to focus on getting through the town!

Default Code

```
1
```

Loopstraction Solution

```
1 for i in range(3):
2     hero.moveRight(1)
3     hero.moveDown(1)
4
5 mouse.moveRight(2)
6
7 for i in range(4):
8     mouse.moveDown(1)
9     mouse.moveRight(1)
10
11 hero.moveRight(3)
```

#42. Long Step Stair

Level Overview and Solutions

Long Step Stair

Intro

This Rift is one big maze! I sure hope we get to the center soon.

Default Code

```
1 mouseDistance = 1
```

Long Step Stair Solution

```
1 mouseDistance = 1
2
3 for i in range(3):
4     hero.moveRight()
5     hero.moveUp()
6
7 hero.jumpRight()
8
9 for i in range(3):
10     mouse.moveUp()
11     mouse.moveRight(mouseDistance)
12     mouseDistance += 1
13
14 hero.moveRight()
```

#43. Plus Minus Bridges

Level Overview and Solutions

Plus Minus Bridges

Intro

What's with all the rivers? It's like something's trying to slow us down or something.

Default Code

```
1 heroDistance = 3
2 mouseDistance = 3
```

Plus Minus Bridges Solution

```
1 heroDistance = 3
2 mouseDistance = 3
3
4 for i in range(3):
5     hero.moveRight(heroDistance)
6     hero.moveUp()
7     heroDistance = heroDistance - 1
8
9 hero.jumpUp()
10
11 for i in range(3):
12     mouse.moveDown(mouseDistance)
13     mouse.moveRight()
14     mouse.moveUp(mouseDistance)
15     mouse.moveRight()
16     mouseDistance = mouseDistance - 1
17
18 mouse.moveRight(1)
19 hero.moveRight()
```

#44. Eye of the Storm

Level Overview and Solutions

Eye of the Storm

Intro

We're close to the center of the Rift. I can feel it!

Default Code

1

Eye of the Storm Solution

```
1 heroWalkDistance = 3
2 mouseWalkDistance = 1
3
4 for i in range(3):
5     hero.moveUp(heroWalkDistance)
6     hero.moveRight()
7     hero.moveDown(heroWalkDistance)
8     hero.moveRight()
9     heroWalkDistance = heroWalkDistance - 1
10
11 hero.moveRight()
12
13 for j in range(4):
14     mouse.moveDown(mouseWalkDistance)
15     mouse.moveRight()
16     mouse.moveUp(mouseWalkDistance)
17     mouse.moveRight()
18     mouseWalkDistance = mouseWalkDistance + 1
19
20 mouse.moveDown(6)
21 hero.moveRight()
```

#48. For Mouse

Level Overview and Solutions

For Mouse

Intro

The Time Storms took you back into the past. Mouse has to be here somewhere.

Default Code

```
1
```

For Mouse Solution

```
1 for i in range(3):
2     hero.moveRight()
3     hero.moveUp(2)
4
5 hero.moveRight()
6 hero.moveDown(6)
7
8 for i in range(3):
9     hero.moveRight()
10    hero.moveUp(2)
11
12 hero.moveRight()
13 hero.moveDown(6)
```

#54. For Inside

Level Overview and Solutions

For Inside

Intro

Okay, this is pretty dangerous stuff. Let's see where you ended up.

Default Code

```
1 for i in range(2):
2     for j in range(3):
3         hero.moveRight()
4         hero.moveDown(2)
5     # Add more actions here
```

For Inside Solution

```
1 for i in range(2):
2     for j in range(3):
3         hero.moveRight()
4         hero.moveDown(2)
5     # Add more actions here
6     hero.moveRight()
7     hero.moveUp(6)
```

#55. Triple Slalom

Level Overview and Solutions

Triple Slalom

Intro

Okay, this place is in ruins. Let's try and find Mouse and get out of here.

Default Code

```
1 for i in range(3):
2     hero.moveRight()
3     # Add an inner loop
4     # Add actions after the inner loop
```

Triple Slalom Solution

```
1 for i in range(3):
2     hero.moveRight()
3     # Add an inner loop
4     # Add actions after the inner loop
5     for i in range(3):
6         hero.moveRight()
7         hero.moveDown()
8         hero.moveLeft()
9         hero.moveDown()
10    hero.moveRight(2)
11    hero.moveUp(6)
```

#56. Serpentes

Level Overview and Solutions

Serpentes

Intro

We should be closer to Mouse now. Let's see where we ended up.

Default Code

1

Serpentes Solution

```
1 for i in range(3):
2     for j in range(3):
3         hero.moveRight()
4         hero.moveUp()
5         hero.moveLeft()
6         hero.moveUp()
7     hero.moveRight(2)
8     hero.moveDown(6)
9     hero.moveRight()
```

#57. Haven Stairs

Level Overview and Solutions

Haven Stairs

Intro

Where are we? Are we back in Haven? I think the better question is "when are we?"

Default Code

```
1 for i in range(2):
2     walkDistance = 1
3     for j in range(3):
4         hero.moveUp(walkDistance)
5         hero.moveRight()
```

Haven Stairs Solution

```
1 for i in range(2):
2     walkDistance = 1
3     for j in range(3):
4         hero.moveUp(walkDistance)
5         hero.moveRight()
6         walkDistance += 1
7     hero.moveDown(6)
8     hero.moveRight()
```

#58. Dark Downstairs

Level Overview and Solutions

Dark Downstairs

Intro

Wait, is that the Darkness? Is this the future or the past?

Default Code

1

Dark Downstairs Solution

```
1 for i in range(2):
2     walkDistance = 3
3
4     for j in range(3):
5         hero.moveDown(walkDistance)
6         hero.moveRight()
7         walkDistance = walkDistance - 1
8     hero.moveUp(6)
9     hero.moveRight()
```

#60. Blind Steps

Level Overview and Solutions

Blind Steps

Intro

What is this, some kind of time fog? The time totem might help clear the way...

Default Code

```
1 hero.moveUp(2)
```

Blind Steps Solution

```
1 hero.moveUp(2)
2 hero.sneakRight(2)
3 hero.jumpRight()
4 hero.moveRight(1)
```

#61. A Loop in the Fog

Level Overview and Solutions

A Loop in the Fog

Intro

We have to keep following Mouse until he stops disappearing!

Default Code

```
1
```

A Loop in the Fog Solution

```
1 for t in range(3):  
2     hero.sneakRight()  
3     hero.sneakUp()  
4  
5 hero.sneakRight(1)  
6 hero.jumpRight()
```

#64. Double Choice

Level Overview and Solutions

Double Choice

Intro

The time fog changes the land underneath it, so you need to be careful.

Default Code

```
1 hero.moveRight()
2
3 if hero.canMoveUp():
4     hero.moveUp()
5     hero.moveRight()
6 else:
7     hero.moveRight()
8     hero.moveUp()
```

Double Choice Solution

```
1 hero.moveRight()
2
3 if hero.canMoveUp():
4     hero.moveUp()
5     hero.moveRight()
6 else:
7     hero.moveRight()
8     hero.moveUp()
9
10 if hero.canMoveUp():
11     hero.moveUp()
12     hero.moveRight()
13 else:
14     hero.moveRight()
15     hero.moveUp()
16
17 hero.moveUp()
18 hero.jumpRight()
19 hero.moveRight()
```


#65. Into the Unknown

Level Overview and Solutions

Into the Unknown

Intro

The time fog is expanding! Be careful out there!

Default Code

```
1 hero.moveUp(2)
2 hero.moveRight(1)
3
4 if hero.canMoveUp():
5     hero.moveUp()
6 else:
7     hero.moveRight()
```

Into the Unknown Solution

```
1 hero.moveUp(2)
2 hero.moveRight()
3
4 if hero.canMoveUp():
5     hero.moveUp()
6 else:
7     hero.moveRight()
8
9 if hero.canMoveUp():
10    hero.moveUp()
11 else:
12    hero.moveRight()
13
14 if hero.canMoveUp():
15    hero.moveUp()
16 else:
17    hero.moveRight()
18
19 if hero.canMoveUp():
20    hero.moveUp()
21 else:
22    hero.moveRight()
23
24
25 hero.jumpRight()
```

#69. Storm Rising

Level Overview and Solutions

Storm Rising

Intro

It's time for you to use what you know about loops and conditionals to find Mouse!

Default Code

```
1 hero.moveRight()
2
3 for i in range(7):
4     if hero.canMoveDown():
5         hero.moveDown()
6     # ELSE if you can't move down, then move RIGHT
```

Storm Rising Solution

```
1 hero.moveRight()
2
3 for i in range(7):
4     if hero.canMoveDown():
5         hero.moveDown()
6     else:
7         hero.moveRight()
8
9 hero.jumpRight()
```

#70. Creeping Through the Tall Grass

Level Overview and Solutions

Creeping Through the Tall Grass

Intro

If you can get to Mouse, maybe this time travel nightmare will end.

Default Code

1

Creeping Through the Tall Grass Solution

```
1 hero.sneakRight(1)
2 for i in range(7):
3     if hero.canMoveUp():
4         hero.sneakUp(1)
5     else:
6         hero.sneakRight(1)
```

#71. Out of the Loop

Level Overview and Solutions

Out of the Loop

Intro

I think that's the real Mouse! We can catch him this time!

Default Code

1

Out of the Loop Solution

```
1 for i in range(10):
2     if hero.canMoveLeft():
3         hero.sneakLeft()
4     else:
5         hero.sneakDown()
```

#74. Right Door

Level Overview and Solutions

Right Door

Intro

You're in the outskirts and need to get into the city. Make sure none of the Quetzals sees you.

Default Code

```
1 enemy = hero.findNearestEnemy()  
2 distance = hero.getDistanceTo(enemy)
```

Right Door Solution

```
1 enemy = hero.findNearestEnemy()  
2 distance = hero.getDistanceTo(enemy)  
3  
4 hero.moveRight(2)  
5 hero.moveUp(distance)  
6 hero.moveRight(4)  
7 hero.moveDown(distance)  
8  
9 mouse.moveRight()  
10 mouse.digRight()  
11 mouse.moveRight(2)
```

#82. Knock Knock

Level Overview and Solutions

Knock Knock

Intro

We're in the outskirts, but be careful. There's no grass to sneak through, so you'll have to use your `hide` command!

Default Code

```
1 hero.hide()
2 mouse.moveRight()
3
4 enemy = hero.findNearestEnemy()
5 enemyDistance = hero.getDistanceTo(enemy)
6
7 while enemyDistance < 3:
8     enemyDistance = hero.getDistanceTo(enemy)
9
10 hero.moveRight(4)
```

Knock Knock Solution

```
1 hero.hide()
2 mouse.moveRight()
3
4 enemy = hero.findNearestEnemy()
5 enemyDistance = hero.getDistanceTo(enemy)
6
7 while enemyDistance < 3:
8     enemyDistance = hero.getDistanceTo(enemy)
9
10 hero.moveRight(4)
11
12 mouse.moveUp()
13 mouse.digRight()
14 mouse.moveRight()
15
16 mouse.digRight()
17 mouse.moveDown()
18
19 hero.moveRight(2)
20 mouse.moveDown(3)
```

#83. Patience

Level Overview and Solutions

Patience

Intro

There are more Quetzal as we get closer to the city. Be sure to hide and move carefully.

Default Code

```
1 hero.hide()
2
3 enemy = hero.findNearestEnemy()
4 distance = hero.getDistanceTo(enemy)
5 while distance < 3:
6     distance = hero.getDistanceTo(enemy)
```

Patience Solution

```
1 hero.hide()
2
3 enemy = hero.findNearestEnemy()
4 distance = hero.getDistanceTo(enemy)
5 while distance < 3:
6     distance = hero.getDistanceTo(enemy)
7
8 hero.moveRight(3)
9 hero.hide()
10 enemy = hero.findNearestEnemy()
11 distance = hero.getDistanceTo(enemy)
12 while distance < 3:
13     distance = hero.getDistanceTo(enemy)
14
15 hero.moveRight(3)
16 mouse.digRight()
17 mouse.moveRight(3)
18 mouse.digRight()
19 mouse.moveDown(3)
```

#84. While Twins

Level Overview and Solutions

While Twins

Intro

The path is getting more complicated. You and Mouse need to help each other move to the exit markers.

Default Code

```
1 hero.hide()
2
3 enemy = hero.findNearestEnemy()
4 distance = hero.getDistanceTo(enemy)
```

While Twins Solution

```
1 hero.hide()
2
3 enemy = hero.findNearestEnemy()
4 distance = hero.getDistanceTo(enemy)
5 while distance < 3:
6     distance = hero.getDistanceTo(enemy)
7
8 hero.moveRight(2)
9 hero.hide()
10
11 mouse.moveRight(4)
12 mouse.digRight()
13 mouse.moveDown()
14
15 distance = hero.getDistanceTo(enemy)
16 while distance < 3:
17     distance = hero.getDistanceTo(enemy)
18
19 hero.moveRight(3)
20 mouse.moveDown(2)
```

#85. Shadow Behind

Level Overview and Solutions

Shadow Behind

Intro

We're right outside the city. You'll need to use that `while` loop to get to the secret entrance.

Default Code

```
1 hero.hide()
```

Shadow Behind Solution

```
1 hero.hide()
2 enemy = hero.findNearestEnemy()
3 enemyDistance = hero.getDistanceTo(enemy)
4 while enemyDistance < 3:
5     enemyDistance = hero.getDistanceTo(enemy)
6
7 hero.moveRight(3)
8 hero.hide()
9
10 for i in range(3):
11     mouse.moveRight()
12     mouse.moveDown()
13     mouse.moveRight()
14     mouse.moveUp()
15
16 mouse.moveRight()
17 mouse.moveDown(4)
18
19 enemyDistance = hero.getDistanceTo(enemy)
20 while enemyDistance < 3:
21     enemyDistance = hero.getDistanceTo(enemy)
22
23 hero.moveRight(3)
```

#90. Be Slow, Be Quick

Level Overview and Solutions

Be Slow, Be Quick

Intro

The Quetzals are continuing to patrol! Use what you know about counter variables to simplify getting around them.

Default Code

```
1 steps = 0
2 hero.hide()
3 while steps < 6:
4     enemy = hero.findNearestEnemy()
5     distance = hero.getDistanceTo(enemy)
6     if distance > 2:
7         hero.moveRight()
8         hero.hide()
9         # Increase steps by 1
```

Be Slow, Be Quick Solution

```
1 steps = 0
2 hero.hide()
3 while steps < 6:
4     enemy = hero.findNearestEnemy()
5     distance = hero.getDistanceTo(enemy)
6     if distance > 2:
7         hero.moveRight()
8         hero.hide()
9         # Increase steps by 1
10        steps = steps + 1
11
12 hero.moveUp()
13 hero.jumpUp()
14 hero.moveUp()
```

#91. One Step Up

Level Overview and Solutions

One Step Up

Intro

There are more guards here. You're getting close to the center of town!

Default Code

```
1 steps = 0
2 hero.hide()
3 while steps < 5:
4     enemy = hero.findNearestEnemy()
5     distance = hero.getDistanceTo(enemy)
```

One Step Up Solution

```
1 steps = 0
2 hero.hide()
3 while steps < 5:
4     enemy = hero.findNearestEnemy()
5     distance = hero.getDistanceTo(enemy)
6     if distance > 2:
7         hero.moveUp()
8         hero.hide()
9         steps += 1
10
11 hero.moveRight()
12 hero.jumpRight()
13 hero.moveRight(2)
14 hero.jumpDown()
15 hero.jumpDown()
16 hero.moveDown()
```

#92. Charge the Time

Level Overview and Solutions

Charge the Time

Intro

The time pedestal is here! Get past these final guards and you can make the time storms go away.

Default Code

1

Charge the Time Solution

```
1 steps = 0
2
3 while steps < 4:
4     enemy = hero.findNearestEnemy()
5     distance = hero.getDistanceTo(enemy)
6     if distance > 2:
7         hero.moveRight()
8         steps += 1
9
10 hero.jumpRight()
11
12 steps = 0
13
14 enemy = hero.findNearestEnemy()
15 while steps < 5:
16     distance = hero.getDistanceTo(enemy)
17     if distance > 2:
18         hero.moveUp()
19         hero.hide()
20         steps += 1
21
22 hero.jumpLeft()
23 hero.moveLeft(2)
24
25 hero.use("pedestal")
```


#95. Lost in Time

Level Overview and Solutions

Lost in Time

Intro

You need to head into the time storm to rescue the citizens of Haven. I'll help you through the Acodus when I can.

Default Code

1

Lost in Time Solution

```
1 for i in range(4):  
2     hero.moveRight()  
3     hero.moveUp()  
4  
5 hero.moveRight(3)
```

#99. Time Call

Level Overview and Solutions

Time Call

Intro

The time storms keep changing things. You can use your time totem to clear your path.

Default Code

```
1 steps = 0
2 while steps < 6:
3     if hero.canMoveRight():
4         hero.moveRight()
5         steps = steps + 1
6     else:
7         hero.castTimeReset()
```

Time Call Solution

```
1 steps = 0
2 while steps < 6:
3     if hero.canMoveRight():
4         hero.moveRight()
5         steps += 1
6     else:
7         hero.castTimeReset()
8
9 hero.moveDown(3)
10
11 for i in range(3):
12     hero.moveLeft()
13     hero.moveDown()
14     hero.moveLeft()
15     hero.moveUp()
```

#100. Time Swap

Level Overview and Solutions

Time Swap

Intro

There's more people to rescue. Use the `while` loop and the **counter variable** to get through the time storm.

Default Code

```
1 actions = 0
2 while actions < 3:
3     if hero.canMoveRight():
4         hero.moveRight(2)
5         actions = actions + 1
6     else:
7         hero.castTimeReset()
```

Time Swap Solution

```
1 actions = 0
2 while actions < 3:
3     if hero.canMoveRight():
4         hero.moveRight(2)
5         actions = actions + 1
6     else:
7         hero.castTimeReset()
8
9 hero.moveUp(2)
10
11 actions = 0
12 while actions < 5:
13     if hero.canMoveLeft():
14         hero.moveLeft()
15         actions = actions + 1
16     else:
17         hero.castTimeReset()
18
19 hero.moveUp(2)
20 hero.moveRight(5)
```

#101. Tempus Fugit

Level Overview and Solutions

Tempus Fugit

Intro

I think you're almost done. Although who's that guy that keeps following you?

Default Code

1

Tempus Fugit Solution

```
1 actions = 4
2 while actions > 0:
3     if hero.canMoveRight():
4         hero.moveRight()
5         hero.moveUp()
6         actions = actions - 1
7     else:
8         hero.castTimeReset(hero)
9
10 hero.jumpRight()
11 hero.moveDown(2)
12 hero.moveLeft()
13
14 for i in range(3):
15     hero.moveLeft()
16     hero.moveDown()
```

#102. Tempus Impetus

Level Overview and Solutions

Tempus Impetus

Intro

This is the last person to rescue. But wait... that's the guy who's been following you!

Default Code

1

Tempus Impetus Solution

```
1 for i in range(3):
2     hero.moveDown()
3     hero.moveLeft()
4
5 hero.moveDown()
6
7 steps = 0
8 while steps < 4:
9     if hero.canMoveRight():
10         hero.moveRight()
11         steps = steps + 1
12     else:
13         hero.castTimeReset(hero)
14
15 while steps > 0:
16     if hero.canMoveUp():
17         hero.moveUp()
18         steps = steps - 1
19     else:
20         hero.castTimeReset(hero)
21
22 hero.moveRight()
23 hero.moveDown(3)
24 hero.moveRight()
```


#110. Dangerous Skies

Level Overview and Solutions

Use for and while loops to create an obstacle course that changes based on the player's performance.

Intro

Create an exciting obstacle course to defeat the Darkness!

Default Code

```
1 avatar = game.createAvatar("dragon") #change ths to "phoenix" if you want
2
3 # define an obstacle type here
4
5 # spawn that obstacle here
```

Dangerous Skies Solution

```
1 avatar = game.createAvatar("phoenix")
2 game.addTextDirections("Ruins are bad, Gems are good.")
3
4 ruin = game.createObstacle("ruin")
5
6 gem = game.createResource("gem")
7 gem.setColor("red")
8
9 for i in range(3):
10     game.spawnTopLane(ruin)
11     game.spawnMiddleLane(gem)
12     game.spawnBottomLane(ruin)
13
14 avatar.setSpeed(30)
15
16 if avatar.getHP() < 5:
17     game.spawnMiddleLane(gem)
18
19 for i in range(2):
20     game.spawnBottomLane(gem)
21     for j in range(5):
22         game.spawnMiddleLane(ruin)
23
24 for i in range(4):
25     choice = game.randomInt(3)
26     if choice == 0:
27         game.spawnTopLane(ruin)
28     if choice == 1:
29         game.spawnMiddleLane(ruin)
30     if choice == 2:
31         game.spawnBottomLane(ruin)
32
33 while avatar.getScore() < 100:
34     game.spawnBottomLane(gem)
35
36 game.win()
```
